

1) $\langle \cdot \rangle$ sk. souč. na $\mathbb{R}^2$ l.ř. báze $B = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right)$ je vzhledem k němu ortonormální

TVRZENÍ 8.50. Je-li V vektorový prostor se skalárním součinem $\langle \cdot, \cdot \rangle$, $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ jeho ortonormální báze, a $\mathbf{u}, \mathbf{w} \in V$, pak

$$\langle \mathbf{u}, \mathbf{w} \rangle = [\mathbf{u}]_B^* [\mathbf{w}]_B.$$

jsme v $\mathbb{R}$, proto * odpovídá T

$$\Rightarrow \left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}_B, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}_B \right\rangle = \left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}_B \right]^* \left[\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}_B \right]$$

$$\left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}_B \right] = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = a_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + a_2 \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

$$x_1 = a_1 + 3a_2 \Rightarrow a_1 = x_1 - 3a_2$$

$$x_2 = 2a_1 + 7a_2 \Rightarrow x_2 = 2x_1 - 6a_2 + 7a_2 \Rightarrow a_2 = x_2 - 2x_1$$

$$\begin{aligned} a_1 &= x_1 - 3x_2 + 6x_1 \\ &\Rightarrow a_1 = 7x_1 - 3x_2 \end{aligned}$$

$$\Rightarrow \left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}_B \right]^T = \begin{pmatrix} 7x_1 - 3x_2 \\ -2x_1 + x_2 \end{pmatrix}^T = (7x_1 - 3x_2, -2x_1 + x_2)$$

$$\left[\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}_B \right] = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = b_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + b_2 \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

$$y_1 = b_1 + 3b_2 \Rightarrow b_1 = y_1 - 3b_2$$

$$y_2 = 2b_1 + 7b_2 \Rightarrow y_2 = 2y_1 - 6b_2 + 7b_2 \Rightarrow b_2 = y_2 - 2y_1$$

$$\Rightarrow b_1 = y_1 - 3y_2 + 6y_1$$

$$b_1 = 7y_1 - 3y_2$$

$$\Rightarrow \left[\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}_B \right] = \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix}$$

$$\left[\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}_B \right]^* \left[\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}_B \right] = (7x_1 - 3x_2, -2x_1 + x_2) \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix} =$$

$$= 49x_1y_1 - 21x_1y_2 - 21x_2y_1 + 9x_2y_2 + 4x_1y_1 - 2x_1y_2 - 2x_2y_1 + x_2y_2 = 53x_1y_1 - 23x_1y_2 - 23x_2y_1 + 10x_2y_2$$